

SUBTITLE GENERATOR SYSTEM

NAME: VINIT SHAH

PRN: 25030142025

SEM: 2

PROGRAMME: MSc CA

PROJECT: JAVA

INTRODUCTION

- SUBTITLES IMPROVE ACCESSIBILITY OF MULTIMEDIA CONTENT
- MANUAL SUBTITLE CREATION IS TIME CONSUMING
- AUTOMATIC SPEECH RECOGNITION (ASR) CAN GENERATE SUBTITLES
- THIS PROJECT AUTOMATES SUBTITLE GENERATION AND TRANSLATION

PROBLEM STATEMENT

- CREATING SUBTITLES MANUALLY REQUIRES SIGNIFICANT TIME
- MANY VIDEOS LACK SUBTITLES FOR GLOBAL AUDIENCES
- NEED FOR AUTOMATED TRANSCRIPTION AND TRANSLATION
- A SCALABLE SYSTEM IS REQUIRED FOR SUBTITLE GENERATION

OBJECTIVES

- GENERATE SUBTITLES AUTOMATICALLY FROM AUDIO/VIDEO
- USE AI BASED SPEECH RECOGNITION FOR TRANSCRIPTION
- TRANSLATE SUBTITLES TO ENGLISH
- PROVIDE DOWNLOADABLE SRT SUBTITLE FILES
- STORE SUBTITLE METADATA FOR ANALYSIS

LITERATURE REVIEW

EXISTING TOOLS: KAPWING, VEED.IO, DESCRIPT

- AUTOMATIC SUBTITLE GENERATION USES AUTOMATIC SPEECH RECOGNITION (ASR) TO CONVERT SPEECH INTO TEXT WITH TIMESTAMPS. [1]
- RECENT DEEP LEARNING MODELS SUCH AS WHISPER SIGNIFICANTLY IMPROVE MULTILINGUAL TRANSCRIPTION ACCURACY. [2]
- STUDIES SHOW SUBTITLE GENERATION SYSTEMS COMBINE SPEECH RECOGNITION, NLP, AND MACHINE TRANSLATION TO AUTOMATE CAPTION GENERATION. [3]
- CURRENT SYSTEMS ACHIEVE 90–97% ACCURACY ON CLEAR SPEECH, BUT PERFORMANCE DEPENDS ON AUDIO QUALITY. [4]

COMPARISON WITH EXISTING TOOLS

Feature	Existing Tools (Kapwing, VEED, Descript)	My Project
Automatic Speech Recognition	Yes	Yes
Subtitle Translation	Limited / Paid	Integrated Azure Translator
Custom Cloud Deployment	No	Yes
Data Storage & Analytics	No	Yes
API-based Modular Architecture	Limited	Yes
Customizable Workflow	No	Yes

NOVELTY & CONTRIBUTION

1. HYBRID SYSTEM ARCHITECTURE

- INTEGRATION OF JAVA WEB UI, PYTHON FASTAPI SERVER, AND AZURE CLOUD SERVICES
- ENABLES EFFICIENT PROCESSING BY SEPARATING WEB SERVICES AND AI PROCESSING MODULES.

2. INTEGRATION OF AI-BASED ASR AND CLOUD TRANSLATION

- USES FASTER-WHISPER FOR MULTILINGUAL SPEECH RECOGNITION.
- USES AZURE AI TRANSLATOR FOR SUBTITLE TRANSLATION INTO ENGLISH.

3. MODULAR CLOUD-BASED DESIGN

- SYSTEM COMPONENTS OPERATE INDEPENDENTLY AND COMMUNICATE VIA APIs.
- IMPROVES SCALABILITY AND MAINTAINABILITY.

4. SUBTITLE DATA ANALYTICS

- STORES FILE METADATA, SOURCE SUBTITLES, AND TRANSLATED SUBTITLES IN AZURE POSTGRESQL.
- ENABLES FUTURE ANALYTICS AND SYSTEM PERFORMANCE EVALUATION.

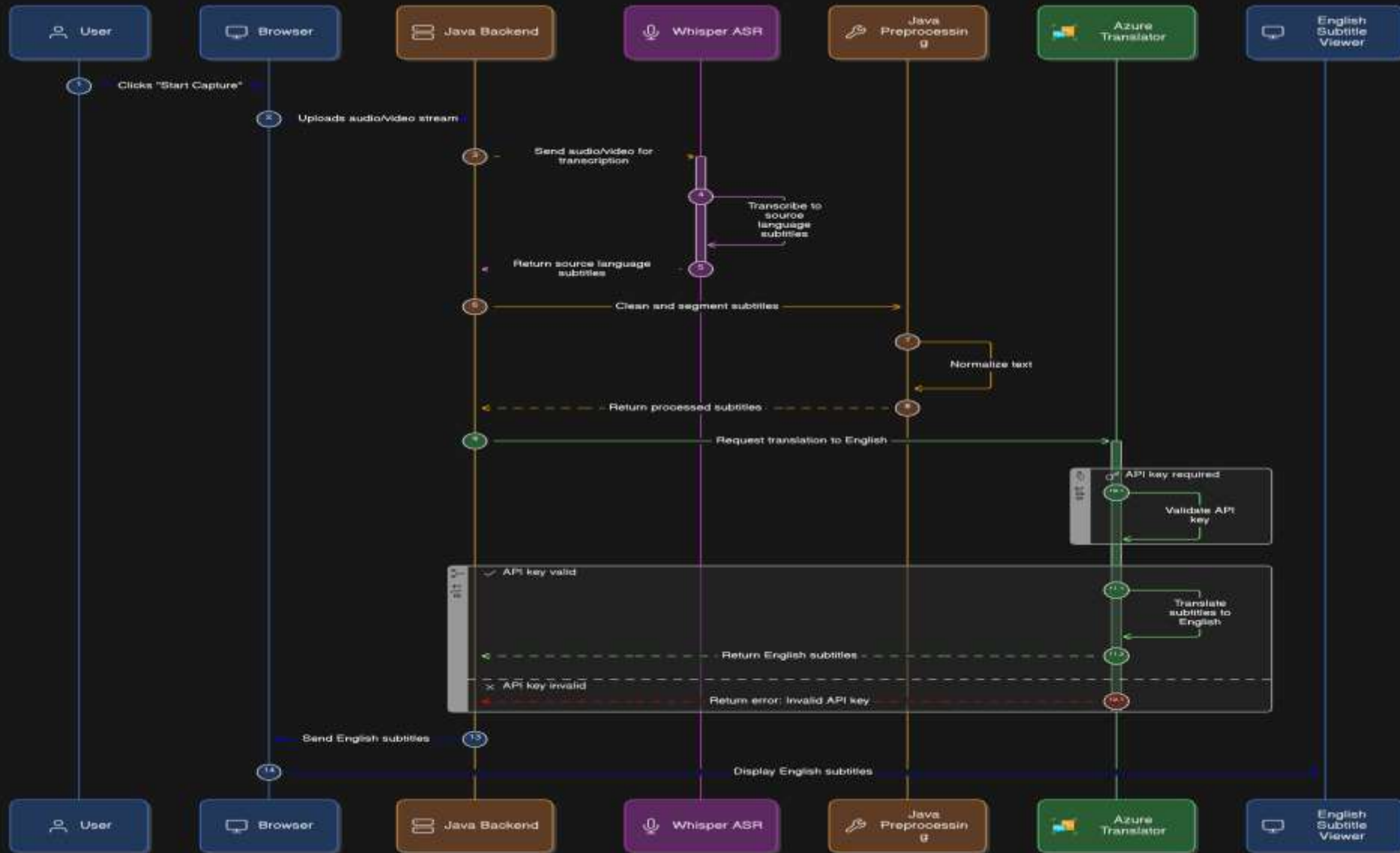
USER REQUIREMENTS

- 1) UPLOAD AUDIO/VIDEO FILE
- 2) SELECT SOURCE LANGUAGE
- 3) VIEW GENERATED ENGLISH SUBTITLES
- 4) DOWNLOAD SUBTITLES AS .SRT
- 5) FAST AND ACCURATE SUBTITLE GENERATION

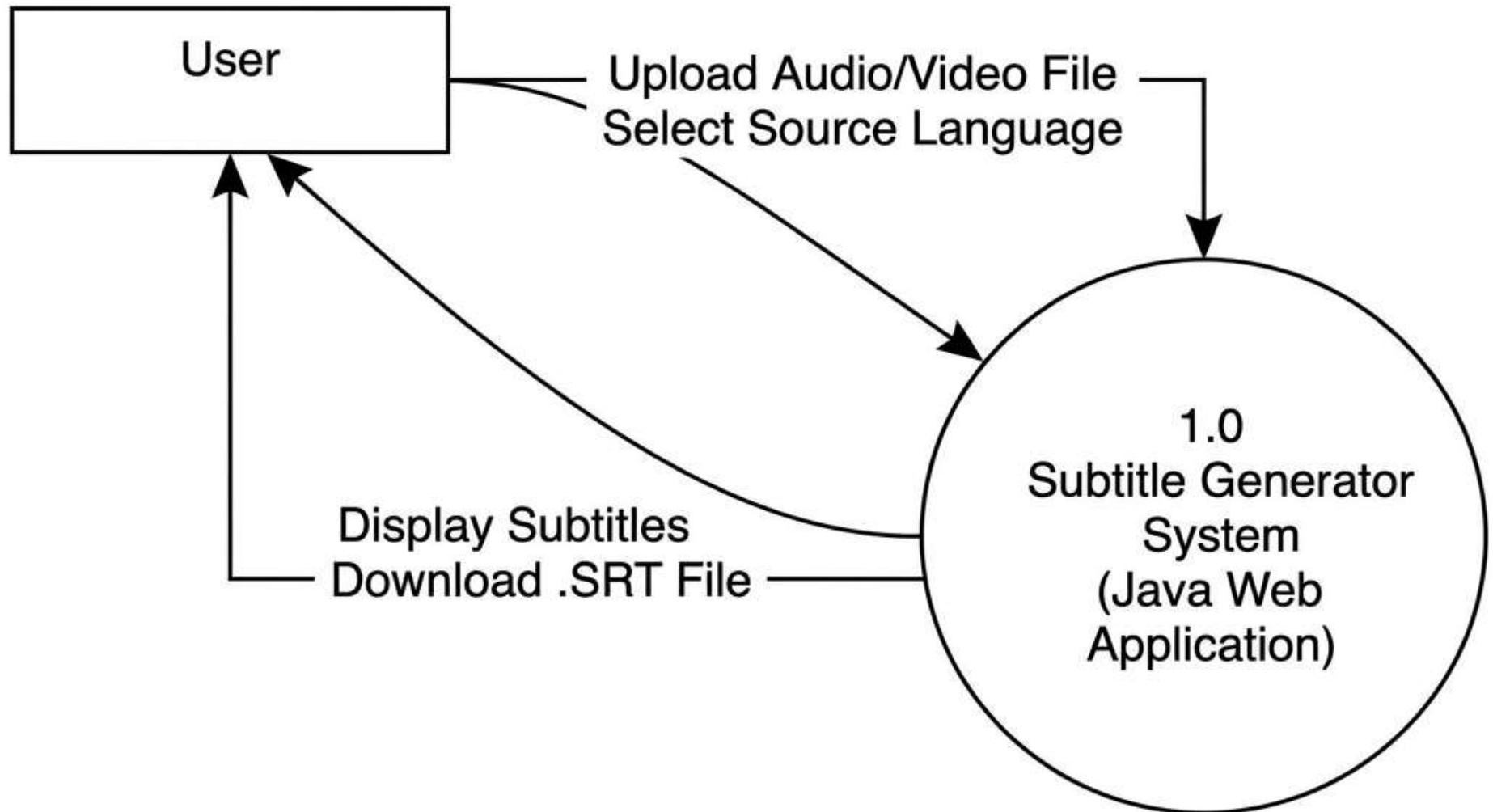
TECHNOLOGY STACK USED

- FRONTEND: JAVA WEB UI
- BACKEND: JSP/SERVLET
- ASR SERVER: PYTHON FASTAPI
- SPEECH MODEL: FASTER WHISPER
- CLOUD: MICROSOFT AZURE
- DATABASE: AZURE POSTGRESQL
- AUDIO PROCESSING: FFMPEG

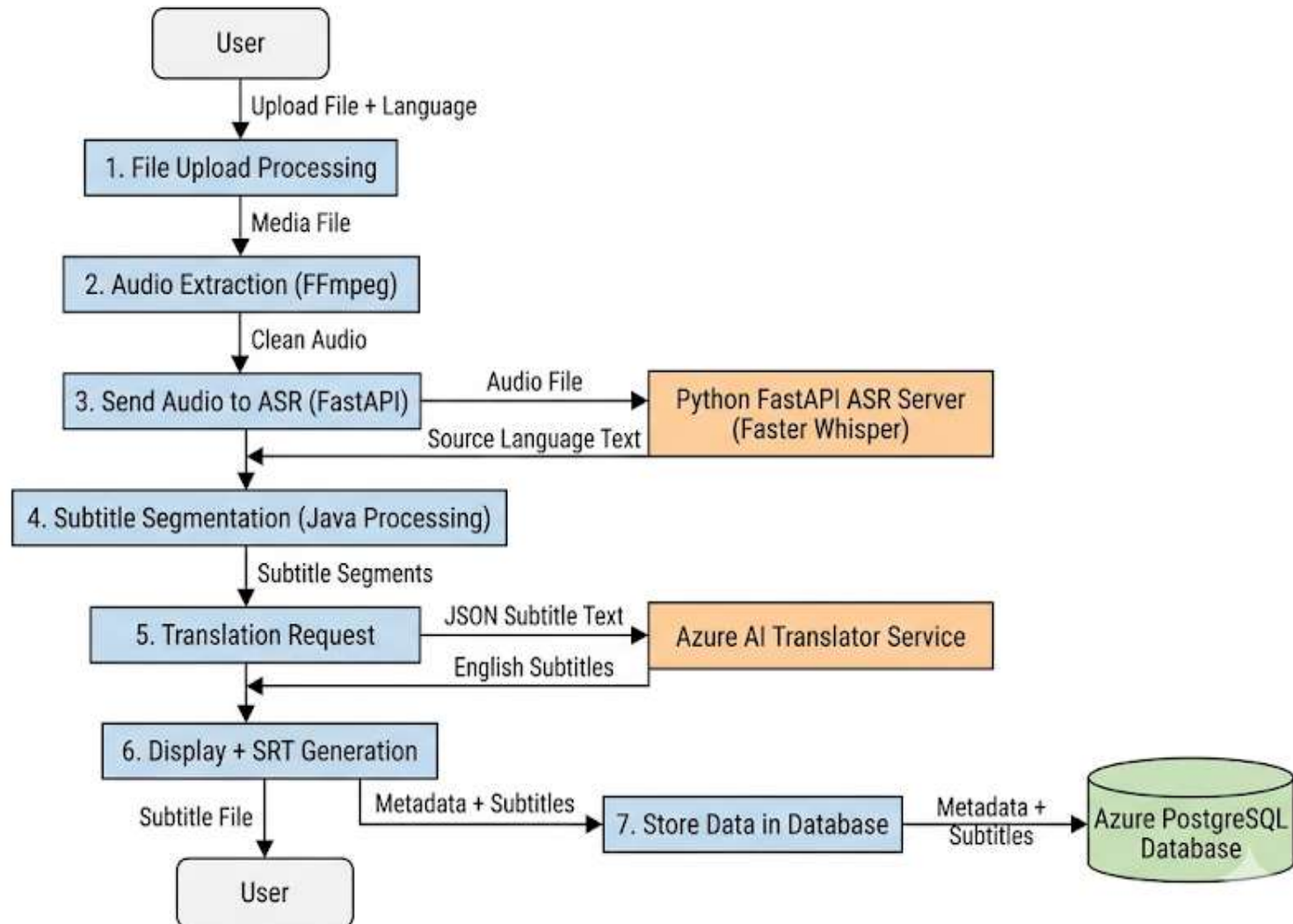
SYSTEM WORKFLOW



DATA FLOW DIAGRAM (0 LEVEL)

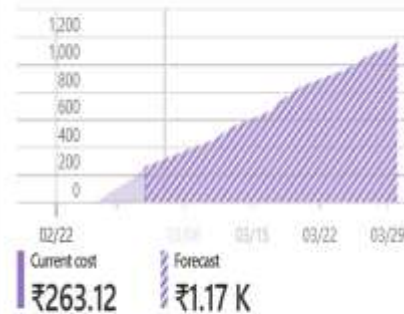


DATA FLOW DIAGRAM (1 LEVEL)



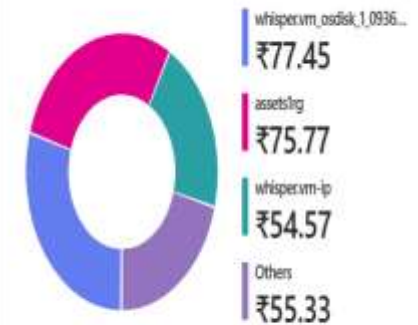
COST ANALYSIS

Spending rate and forecast



[View details](#)

Costs by resource



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Resource	Usage Example	Estimated Cost
Azure VM	6 hours/day usage	~\$34/month (\$140/month)
PostgreSQL	Basic Flexible Server	~\$50/month (\$75/month)
Azure Translator	200K characters/month	~\$2 (\$10 per 1M chars)
Total Estimated Monthly Cost		\$86/month

Component	Cloud Service	Service Model	Purpose	Example Configuration	Approx Cost
Speech Recognition (Whisper ASR)	Azure Virtual Machine	IaaS	Hosts Python Whisper server for speech-to-text processing	Standard D4s v3 (4 vCPU, 16GB RAM)	~\$0.19/hour (~\$140/month if running 24/7)
Subtitle Database	Azure Database for PostgreSQL	PaaS	Stores metadata, subtitles, and translations	Flexible Server – 2 vCore, 8GB RAM	~\$50–\$70/month
Text Translation	Azure AI Translator	SaaS	Translates subtitles from source language to English	Translator Text API	~\$10 per 1M characters
Application Backend	Java Servlet (Tomcat)	-	Handles file upload, calls Whisper API, manages translation and DB	Hosted on VM or App Service	Included in VM cost
Frontend	JSP + JavaScript	Web App	Displays subtitles and allows SRT download	Hosted with backend	Included

FUTURE SCOPE

- GENERATE SUBTITLES IN MULTIPLE LANGUAGES OTHER THAN ENGLISH
- WIDER SOURCE LANGUAGE COVERAGE
- FULLY AUTOMATED SUBTITLE GENERATION (NO HITL REQUIRED)

REFERENCES / BIBLIOGRAPHY

- [HTTPS://GITHUB.COM/SYSTRAN/FASTER-WHISPER](https://github.com/SYSTRAN/faster-whisper)
- FASTAPI DOCUMENTATION
- FFMPEG DOCUMENTATION
- AZURE POSTGRESQL DOCUMENTATION
- LITERATURE REVIEW:

[1] [HTTPS://WWW.INTERNATIONALJOURNALSSRG.ORG/IJCSE/PAPER-DETAILS?ID=313](https://www.internationaljournalssrg.org/IJCSE/paper-details?id=313)

[2] [HTTPS://WWW.RESEARCHGATE.NET/PUBLICATION/396973046](https://www.researchgate.net/publication/396973046) AUTOMATED SPEECH-TO-TEXT CAPTIONING FOR VIDEOS AND NOISE ROBUSTNESS ANALYSIS USING OPEN AI WHISPER A PERFORMANCE AND ENHANCEMENT STUDY

[3] [HTTPS://WWW.KAPWING.COM/SUBTITLES](https://www.kapwing.com/subtitles)

[4] [HTTPS://FLUXNOTE.IO/GUIDES/BEST-FREE-AI-SUBTITLE-GENERATORS-2026](https://fluxnote.io/guides/best-free-ai-subtitle-generators-2026)